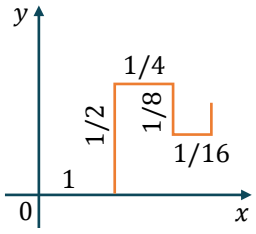
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. A particle begins at the origin and moves successively in the following manner as shown,
1 unit to the right, 1/2 unit up, 1/4 unit to the right,
1/8 unit down, 1/16 unit to the right etc.



The length of each move is half the length of the previous move and movement continues in the $'zigzag'$ manner infinitely. The co-ordinates of the point to which the $'zigzag'$ converges is –
(A) (4/3, 2/3) (B) (4/3,2/5) (C) (3/2,2/3) (D) (2,2/5)

MSL001

2. Consider the points $P(2,-4)$; $Q(4,-2)$ and $R(7,1)$. The points P, Q, R -
(A) form an equilateral triangle
(B) form a right angled triangle
(C) form an isosceles triangle which is not equilateral
(D) are collinear.

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MSL157

3. Suppose ABC is a triangle with 3 acute angles A, B and C. The point whose coordinates are $(\cos B - \sin A, \sin B - \cos A)$ can be in the -
(A) first and 2nd quadrant (B) second the 3rd quadrant
(C) third and 4th quadrant (D) second quadrant only

MSL002

4. If in triangle ABC, $A \equiv (1,10)$, circumcentre $\equiv \left(-\frac{1}{3},\frac{2}{3}\right)$ and orthocentre $\equiv \left(\frac{11}{3},\frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is-
(A) (1,–11/3) (B) (1,5) (C) (1,–3) (D) (1,6)

MSL003

5. The co-ordinates of the orthocentre of the triangle bounded by the lines, $4x - 7y + 10 = 0$; $x + y = 5$ and $7x + 4y = 15$ is-
(A) (2,1) (B) (–1,2) (C) (1,2) (D) (1,–2)

MSL004

6. The length of a line segment AB is 10 units. If the coordinates of one extremity are (2,–3) and the abscissa of the other extremity is 10 then the sum of all possible values of the ordinate of the other extremity is -
(A) 3 (B) –4 (C) 12 (D) –6

MSL158

For More Join: @IITJEE_Advanced

7. A triangle has two of its vertices at (0,1) and (2,2) in the cartesian plane. Its third vertex lies on the x-axis. If the area of the triangle is 2 square units then the sum of the possible abscissae of the third vertex, is-

(A) –4 (B) 0 (C) 5 (D) 6

MSL005

8. A line passes through (2,2) and cuts a triangle of area 9 square units from the first quadrant. The sum of all possible values for the slope of such a line, is-

(A) –2.5 (B) –2 (C) –1.5 (D) –1

MSL006

9. Let $A(2,-3)$ and $B(-2,1)$ be vertices of a ΔABC. If the centroid of ΔABC moves on the line $2x + 3y = 1$, then the locus of the vertex C is-

(A) $2x + 3y = 9$ (B) $2x - 3y = 7$

(C) $3x + 2y = 5$ (D) $3x - 2y = 3$

MSL007

10. A stick of length 10 units rests against the floor and a wall of a room. If the stick begins to slide on the floor then the locus of its middle point is :

(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$

(C) $x^2 + y^2 = 100$ (D) none

MSL008

11. The area of the parallelogram formed by the lines $3x + 4y = 7a$; $3x + 4y = 7b$; $4x + 3y = 7c$ and $4x + 3y = 7d$ is-

(A) $\frac{|(a-b)(c-d)|}{7}$ (B) $|(a-b)(c-d)|$

(C) $\frac{|(a-b)(c-d)|}{49}$ (D) $7|(a-b)(c-d)|$

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MSL009

12. If the lines $\left.\begin{array}{l}\lambda x+(\sin\alpha)y+\cos\alpha=0\\ x+(\cos\alpha)y+\sin\alpha=0\\ x-(\sin\alpha)y+\cos\alpha=0\end{array}\right]$ pass through the same point where $\alpha \in R$ then λ lies in the interval -

(A) [–1,1] (B) $\left[-\sqrt{2},\sqrt{2}\right]$ (C) [–2,2] (D) $(-\infty,\infty)$

MSL010

13. If the straight lines, $ax + amy + 1 = 0$, $bx + (m + 1)$ by $+ 1 = 0$ and $cx + (m + 2)cy + 1 = 0$, $m \neq 0$ are concurrent then a, b, c are in :

(A) A.P. only for $m = 1$ (B) A.P. for all m

(C) G.P. for all m (D) H.P. for all m

MSL011

14. Consider a parallelogram whose sides are represented by the lines $2x + 3y = 0$; $2x + 3y - 5 = 0$; $3x - 4y = 0$ and $3x - 4y = 3$. The equation of the diagonal not passing through the origin, is-

(A) $21x - 11y + 15 = 0$ (B) $9x - 11y + 15 = 0$

(C) $21x - 29y - 15 = 0$ (D) $21x - 11y - 15 = 0$

MSL012

For More Join: @IITJEE_Advanced

15. Let $S = \{(x, y) | \ x^2 + 2xy + y^2 - 3x - 3y + 2 = 0\}$, then S -
(A) consists of two coincident lines.
(B) consists of two parallel lines which are not coincident.
(C) consists of two intersecting lines.
(D) is a parabola.

MSL013

16. The line $x = c$ cuts the triangle with corners (0,0); (1,1) and (9,1) into two region. For the area of the two regions to be the same c must be equal to-
(A) 5/2 (B) 3 (C) 7/2 (D) 3 or 15

MSL014

17. If m and b are real numbers and $mb > 0$, then the line whose equation is $y = mx + b$ cannot contain the point-
(A) (0,2009) (B) (2009,0)
(C) (0,–2009) (D) (20,–100)

MSL015

18. In a triangle ABC, side AB has the equation $2x + 3y = 29$ and the side AC has the equation, $x + 2y = 16$. If the mid- point of BC is (5,6) then the equation of BC is :
(A) $x - y = -1$ (B) $5x - 2y = 13$
(C) $x + y = 11$ (D) $3x - 4y = -9$

MSL016

19. The vertex of the right angle of a right angled triangle lies on the straight line $2x - y - 10 = 0$ and the two other vertices, at points (2,–3) and (4,1) then the area of triangle in sq. units is-
(A) $\sqrt{10}$ (B) 3 (C) $\frac{33}{5}$ (D) 11

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MSL017

20. The co-ordinates of a point P on the line $2x - y + 5 = 0$ such that $|PA - PB|$ is maximum where A is (4,–2) and B is (2,–4) will be :
(A) (11,27) (B) (–11,–17) (C) (–11,17) (D) (0,5)

MSL018

21. The line $(k + 1)^2x + ky - 2k^2 - 2 = 0$ passes through a point regardless of the value k. Which of the following is the line with slope 2 passing through the point?
(A) $y = 2x - 8$ (B) $y = 2x - 5$ (C) $y = 2x - 4$ (D) $y = 2x + 8$

MSL019

22. If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the x-axis then the value of k :
(A) is equal to 1 (B) is equal to –1
(C) is equal to 2 (D) does not exist in the set of real numbers

MSL020

23. Given the points $A(0,4)$ and $B(0,-4)$, the equation of the locus of the point P such that $|AP - BP| = 6$ is -
(A) $9x^2 - 7y^2 + 63 = 0$ (B) $9x^2 - 7y^2 - 63 = 0$
(C) $7x^2 - 9y^2 + 63 = 0$ (D) $7x^2 - 9y^2 - 63 = 0$

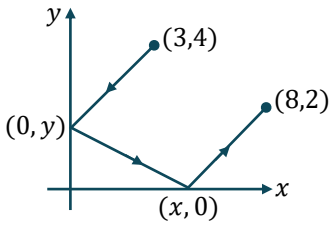
MSL065

For More Join: @IITJEE_Advanced

24. If x_1, y_1 are the roots of $x^2 + 8x - 20 = 0$, x_2, y_2 are the roots of $4x^2 + 32x - 57 = 0$ and x_3, y_3 are the roots of $9x^2 + 72x - 112 = 0$, then the points, (x_1, y_1), (x_2, y_2) and (x_3, y_3) -
(A) are collinear
(B) form an equilateral triangle
(C) form a right angled isosceles triangle
(D) are concyclic

MSL066

25. Suppose that a ray of light leaves the point (3,4), reflects off the y-axis towards the x-axis, reflects off the x-axis, and finally arrives at the point (8,2). The value of x, is -



(A) $x = 4\frac{1}{2}$ (B) $x = 4\frac{1}{3}$ (C) $x = 4\frac{2}{3}$ (D) $x = 5\frac{1}{3}$

MSL067

26. Family of lines represented by the equation $(\cos\theta + \sin\theta)x + (\cos\theta - \sin\theta)y - 3(3\cos\theta + \sin\theta) = 0$ passes through a fixed point M for all real values of θ. The reflection of M in the line $x - y = 0$, is-
(A) (6,3) (B) (3,6) (C) (−6,3) (D) (3,−6)

MSL068

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27. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2, then sum of all possible values of a is-
(A) 17 (B) −19 (C) 19 (D) −17

MSL069

28. A triangle ABC is formed by the lines $2x - 3y - 6 = 0$; $3x - y + 3 = 0$ and $3x + 4y - 12 = 0$. If the points $P(\alpha,0)$ and $Q(0,\beta)$ always lie on or inside the ΔABC, then:
(A) $\alpha \in [-1,2]$ and $\beta \in [-2,3]$ (B) $\alpha \in [-1,3]$ and $\beta \in [-2,4]$
(C) $\alpha \in [-2,4]$ and $\beta \in [-3,4]$ (D) $\alpha \in [-1,3]$ and $\beta \in [-2,3]$

MSL070

29. A and B are any two points on the positive x and y axis respectively satisfying $2(OA) + 3(OB) = 10$. If P is the middle point of AB then the locus of P is-
(A) $2x + 3y = 5$ (B) $2x + 3y = 10$ (C) $3x + 2y = 5$ (D) $3x + 2y = 10$

MSL159

30. m, n are integer with $0 < n < m$. A is the point (m, n) on the cartesian plane. B is the reflection of A in the line $y = x$. C is the reflection of B in the y-axis, D is the reflection of C in the x-axis and E is the reflection of D in the y-axis. The area of the pentagon $ABCDE$ is -
(A) $2m(m + n)$ (B) $m(m + 3n)$ (C) $m(2m + 3n)$ (D) $2m(m + 3n)$

MSL160

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2 - 3|x| + 2 = 0$ and the y-coordinates of the vertices are the roots of the equation $y^2 - 3y + 2 = 0$ then the possible vertices of the square is/are-

(A) (1,1), (2,1), (2,2), (1,2) (B) (–1,1), (–2,1), (–2,2), (–1,2)

(C) (2,1), (1,–1),(1,2),(2,2) (D) (–2,1), (–1,–1),(–1,2), (–2,2)

MSL071

2. The sides of a triangle are the straight lines $x + y = 1$; $7y = x$ and $\sqrt{3}y + x = 0$. Then which of the following is an interior point of the triangle ?

(A) circumcentre (B) centroid (C) incentre (D) orthocentre

MSL161

3. The area of triangle ABC is 20 square units. The co-ordinates of vertex A are (–5,0) and B are (3,0). The vertex C lies on the line, $x - y = 2$. The co-ordinates of C are -

(A) (5,3) (B) (–3,–5) (C) (–5,–7) (D) (7,5)

MSL162

4. Line $\frac{x}{a}+\frac{y}{b}=1$ cuts the co-ordinate axes at $A(a,0)$ and $B(0,b)$ and the line $\frac{x}{a'}+\frac{y}{b'}=-1$ at $A'(-a',0)$ and $B'(0,-b')$. If the points A, B, A', B' are concyclic then the orthocentre of the triangle ABA' is -

(A) (0,0) (B) $(0,b')$ (C) $\left(0,\frac{aa'}{b}\right)$ (D) $\left(0,\frac{bb'}{a}\right)$

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MSL072

5. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines $ax + by + c = 0$ are concurrent at :

(A) (1/2, 3/2) (B) (–1/2, –3/2) (C) (–1/2, 3/2) (D) (1/2, –3/2)

MSL073

6. Equation of a straight line passing through the point (2,3) and inclined at an angle of arc tan $\frac{1}{2}$ with the line $y + 2x = 5$ is-

(A) $y = 3$ (B) $x = 2$

(C) $3x + 4y - 18 = 0$ (D) $4x + 3y - 17 = 0$

MSL163

7. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angles between them is θ. Which of the following relations hold good?

(A) $m_1 + m_2 = 5/4$

(B) $m_1 m_2 = 3/8$

(C) acute angle between L_1 and L_2 is $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$.

(D) sum of the abscissa and ordinate of the point P is –1.

MSL074

For More Join: @IITJEE_Advanced

8. The position of the point (1, 1) w.r.t. the lines $L_1 = 2x + 3y + 7 = 0$ and $L_2 = 4x + 6y - 19 = 0$ is such that :

(A) it is below L_2 (B) it is above L_1

(C) it is between L_1 & L_2 (D) it is above both L_1 & L_2

MSL164

9. Two adjacent sides of a rhombus are $2x + 3y = a - 5$ and $3x + 2y = 4 - 2a$ and its diagonals intersect at the point (1,2), then a can be -

(A) –16 (B) 16 (C) $-\frac{10}{3}$ (D) $\frac{10}{3}$

MSL165

10. If one side of a square is parallel to $3x - 4y = 0$ & its area being 16 while centre being (1, 1), then find equation of sides of square.

(A) $3x - 4y + 11 = 0$ (B) $3x - 4y - 9 = 0$ (C) $4x + 3y + 3 = 0$ (D) $4x + 3y - 17 = 0$

MSL166

11. The equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of lines whose slopes are m and m^2, if angle between the lines is θ , then ($a > 0$)

(A) $\tan\theta = \frac{4}{9}$ (B) $\tan\theta = \frac{2}{9}$ (C) $\tan 2\theta = \frac{18}{65}$ (D) $\tan 2\theta = \frac{36}{77}$

MSL075

12. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.

(A) $2x - y + 3 = 0$ (B) $x + 2y - 6 = 0$ (C) $2x + y - 7 = 0$ (D) $x - 2y - 6 = 0$

MSL076

13. $2x^2 - 3xy + y^2 + 5x - 4y + 3 = 0$ represent two straight lines of slope m_1 and m_2. Then choose correct options -

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(A) harmonic mean of m_1 and m_2 is $\frac{4}{3}$

(B) arithmetic mean of m_1 and m_2 is 3

(C) point of intersection of these lines is (0, –1)

(D) point of intersection of these lines is (–2, –1)

MSL167

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 14 to 16

Consider two points $A \equiv (1,2)$ and $B \equiv (3,-1)$. Let M be a point on the straight line $L \equiv x + y = 0$.

14. If M be a point on the line $L = 0$ such that $AM + BM$ is minimum, then the reflection of M in the line $x = y$ is -

(A) (1,–1) (B) (–1,1) (C) (2,–2) (D) (–2,2)

MSL089

15. If M be a point on the line $L = 0$ such that $|AM - BM|$ is maximum, then the distance of M from $N \equiv (1,1)$ is-

(A) $5\sqrt{2}$ (B) 7 (C) $3\sqrt{5}$ (D) 10

MSL090

16. If M be a point on the line $L = 0$ such that $|AM - BM|$ is minimum, then the area of ΔAMB equals-

(A) $\frac{13}{4}$ (B) $\frac{13}{2}$ (C) $\frac{13}{6}$ (D) $\frac{13}{8}$

MSL091

For More Join: @IITJEE_Advanced

Paragraph for Question Nos. 17 to 18

Let $ABCD$ is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that $AE = AF$. Let P be a point inside the square $ABCD$.

17. The maximum possible area of quadrilateral $CDFE$ is -

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

MSL092

18. The value of $(PA)^2 - (PB)^2 + (PC)^2 - (PD)^2$ can not be equal to -

(A) 3 (B) 2 (C) 1 (D) 0

MSL093

MATCHING LIST TYPE QUESTION

19.

	List-I	**List-II**
(I)	If the incentre of the triangle formed by the coordinate axes and the line $3x + 4y = 2$ is at (a, b), then $a + b$ equals	(P) 1
(II)	The line $2x - 3y = 6$ is shifted by 1 unit closer to the origin, the equation of line in new position is $2x - 3y = 6 + k$, then value of $\|k\|$ is	(Q) 2
(III)	Sum of the abscissa of points on the line $x - y = 3$ which lie at a unit distance from the line $4x - 3y = 12$ is	(R) 3
(IV)	The integral part of 'a' for which the point $(2, a)$ lies between the parallel lines $x + 2y = 4$ and $x + 2y = 5$ is	(S) 6

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Which of the following is correct ?

(A) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ S

(B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ P

(C) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ P

(D) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ Q

MSL094

MATRIX MATCH TYPE QUESTION

20. In ΔABC, let slope of side AB is $\frac{1}{3}$ & internal angle bisector AD is $3x = 4y$. Also mirror image of vertex B w.r.t. line AD is (13, 16) & point D is (12, 9) (where D lies on BC).

	Column-I	**Column-II**
(A)	Co-ordinates of vertex 'B' are (a, b), then $(a - b)$ is equal to	(P) $\frac{13}{9}$
(B)	Slope of side AC, is	(Q) $\frac{25}{4}$
(C)	x-intercept of external angle bisector of vertex 'A' is	(R) 11
(D)	Abscissa of point C is $\frac{a}{b}$ where a & b are coprime numbers, then $\|a - b\|$ is equal to	(S) 13
		(T) 15

MSL095

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. Let $O(0,0), A(6,0)$ and $B(3, \sqrt{3})$ be the vertices of ΔOAB. Let R be the region consisting of all those points P inside ΔOAB which satisfy $d(P,OA) \le$ minimum $\big(d(P,OB), d(P,AB)\big)$ where $d(P,OA)$, $d(P,OB)$ and $d(P,AB)$ represent the distance of P from the sides OA, OB and AB respectively. If the area of region R is $9(a-\sqrt{b})$ where a and b are coprime, then find the value of $(a+b)$

MSL021

2. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle. (In unit)

MSL022

3. The line $3x+2y=24$ meets the y-axis at A & the x-axis at B. The perpendicular bisector of AB meets the line through $(0,-1)$ parallel to x-axis at C. Find the area of the triangle ABC. (In Sq. unit)

MSL023

4. Consider a ΔABC whose sides AB, BC and CA are represented by the straight lines $2x+y=0$, $x+py=q$ and $x-y=3$ respectively. The point P is (2, 3). If P is the circumcentre, then find the value of $(p+q)$.

MSL024

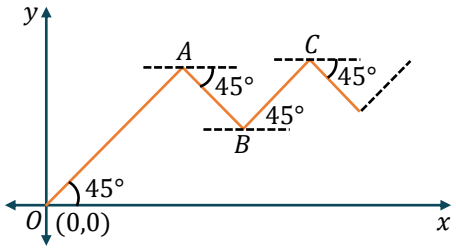
5. The two-line pairs $y^2-4y+3=0$ and $x^2+4xy+4y^2-5x-10y+4=0$ enclose a 4 sided convex polygon find area of the polygon.

MSL025

6. Point O, A, B, C ..... are shown in figure where $OA = 2AB = 4BC =$ ..... so on. Let A be the centroid of a triangle whose orthocentre and circumcentre are $(2,4)$ and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point $O(0,0)$ along the straight line in $zig-zag$ fashion and terminates ultimately at point $P(\alpha, \beta)$ then find the value of $(\alpha+\beta)$

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MSL026



7. If the straight line drawn through the point $P(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x-4y+8=0$ at Q. Find the length PQ. (in unit)

MSL027

8. The sides of a triangle have the combined equation $x^2-3y^2-2xy+8y-4=0$. The third side, which is variable always passes through the point $(-5,-1)$. If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a,b), then find $\left(a+\frac{1}{b^2}\right)$.

MSL028

For More Join: @IITJEE_Advanced

9. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are $A(-8, 5)$; $B(-15, -19)$ and $C(1, -7)$ has the equation $ax + 2y + c = 0$. Find $a + c$.

MSL029

10. If algebraic sum of lengths of perpendiculars drawn from vertices of a hexagon $ABCDEF$, $A(-2, -1)$, $B(0, -2)$, $C(4,0)$, $D(3,2)$, $E(2,3)$, $F(-1,4)$ on a variable line $ax + by + c = 0$ is zero, then variable line always passes through a fixed point (p, q), then the value of $p + q$ is

MSL030

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let the equations of two sides of a triangle be $3x-2y+6=0$ and $4x+5y-20=0$. If the orthocenter of this triangle is at (1,1), then the equation of its third side is: **[JEE (Main) 2019]**

(1) $122y-26x-1675=0$ (2) $26x+61y+1675=0$

(3) $122y+26x+1675=0$ (4) $26x-122y-1675=0$

MSL031

2. Let S be the set of all triangles in the xy-plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50 sq. units, then the number of elements in the set S is: **[JEE (Main) 2019]**

(1) 9 (2) 18 (3) 32 (4) 36

MSL032

3. If the line $3x+4y-24=0$ intersects the x-axis at the point A and the y-axis at the point B, then the incentre of the triangle OAB, where O is the origin, is **[JEE (Main) 2019]**

(1) $(3,4)$ (2) $(2,2)$ (3) $(4,4)$ (4) $(4,3)$

MSL033

4. Two vertices of a triangle are (0,2) and (4,3). If its orthocenter is at the origin, then its third vertex lies in which quadrant? **[JEE (Main) 2019]**

(1) Fourth (2) Second (3) Third (4) First

MSL035

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5. Two sides of a parallelogram are along the lines, $x+y=3$ and $x-y+3=0$. If its diagonals intersect at (2,4), then one of its vertex is: **[JEE (Main) 2019]**

(1) (2,6) (2) (2,1) (3) (3,5) (4) (3,6)

MSL036

6. If a circle of radius R passes through the origin O and intersects the coordinate axes at A and B, then the locus of the foot of perpendicular from O on AB is : **[JEE (Main) 2019]**

(1) $(x^2+y^2)^2=4Rx^2y^2$ (2) $(x^2+y^2)(x+y)=R^2xy$

(3) $(x^2+y^2)^3=4R^2x^2y^2$ (4) $(x^2+y^2)^2=4R^2x^2y^2$

MSL040

7. A point on the straight line, $3x+5y=15$ which is equidistant from the coordinate axes will lie only in: **[JEE (Main) 2019]**

(1) 1st and 2nd quadrants (2) 4th quadrant

(3) 1st, 2nd and 4th quadrant (4) 1st quadrant

MSL041

8. Suppose that the points (h,k), (1,2) and $(-3,4)$ lie on the line L_1. If a line L_2 passing through the points (h,k) and (4,3) is perpendicular to L_1, then $\frac{k}{h}$ equals: **[JEE (Main) 2019]**

(1) 3 (2) $-\frac{1}{7}$ (3) $\frac{1}{3}$ (4) 0

MSL042

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9. Slope of a line passing through $P(2,3)$ and intersecting the line, $x+y=7$ at a distance of 4 units from P, is **[JEE (Main) 2019]**

(1) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$ (2) $\frac{1-\sqrt{5}}{1+\sqrt{5}}$ (3) $\frac{1-\sqrt{7}}{1+\sqrt{7}}$ (4) $\frac{\sqrt{7}-1}{\sqrt{7}+1}$

MSL043

10. A rectangle is inscribed in a circle with a diameter lying along the line $3y=x+7$. If the two adjacent vertices of the rectangle are (–8, 5) and (6, 5), then the area of the rectangle (in sq. units) is :- **[JEE (Main) 2019]**

(1) 72 (2) 84 (3) 98 (4) 56

MSL045

11. The region represented by $|x-y|\le 2$ and $|x+y|\le 2$ is bounded by a: **[JEE (Main) 2019]**

(1) square of side length $2\sqrt{2}$ units (2) rhombus of side length 2 units

(3) square of area 16 sq. units (4) rhombus of area $8\sqrt{2}$ sq. units

MSL046

12. A straight-line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x+y=0$. Then an equation of the line L is: **[JEE (Main) 2019]**

(1) $(\sqrt{3}+1)x+(\sqrt{3}-1)y=8\sqrt{2}$ (2) $(\sqrt{3}-1)x+(\sqrt{3}+1)y=8\sqrt{2}$

(3) $\sqrt{3}x+y=8$ (4) $x+\sqrt{3}y=8$

MSL049

13. The locus of the mid-points of the perpendiculars drawn from points on the line, $x=2y$ to the line $x=y$ is : **[JEE (Main) 2020]**

(1) $2x-3y=0$ (2) $7x-5y=0$ (3) $5x-7y=0$ (4) $3x-2y=0$

MSL051

14. Let $A(1,0), B(6,2)$ and $C\left(\frac{3}{2},6\right)$ be the vertices of a triangle ABC. If P is a point inside the triangle ABC such that the triangles APC, APB and BPC have equal areas, then the length of the line segment PQ, where $Q\left(\frac{-7}{6},\frac{-1}{3}\right)$ is the point, is _____. **[JEE (Main) 2020]**

MSL052

15. Let two points be $A(1,-1)$ and $B(0,2)$. If a point $P(x',y')$ be such that the area of $\Delta PAB=5$ sq. units and it lies on the line, $3x+y-4\lambda=0$, then a value of λ is **[JEE (Main) 2020]**

(1) 1 (2) 4 (3) 3 (4) -3

MSL053

16. Let C be the centroid of the triangle with vertices $(3,-1),(1,3)$ and $(2,4)$. Let P be the point of intersection of the lines $x+3y-1=0$ and $3x-y+1=0$. Then the line passing through the points C and P also passes through the point: **[JEE (Main) 2020]**

(1) $(7,6)$ (2) $(-9,-6)$ (3) $(-9,-7)$ (4) $(9,7)$

MSL054

17. The set of all possible values of θ in the interval $(0,\pi)$ for which the points $(1,2)$ and $(\sin\theta,\cos\theta)$ lie on the same side of the line $x+y=1$ is: **[JEE (Main) 2020]**

(1) $\left(0,\frac{\pi}{4}\right)$ (2) $\left(0,\frac{3\pi}{4}\right)$ (3) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ (4) $\left(0,\frac{\pi}{2}\right)$

MSL055

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18. If the perpendicular bisector of the line segment joining the points $P(1,4)$ and $Q(k,3)$ has y-intercept equal to -4, then a value of k is: - **[JEE (Main) 2020]**

(1) $\sqrt{15}$ (2) -2 (3) $\sqrt{14}$ (4) -4

MSL057

19. If the line, $2x-y+3=0$ is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines $4x-2y+\alpha=0$ and $6x-3y+\beta=0$, respectively, then the sum of all possible values of α and β is. **[JEE (Main) 2020]**

MSL058

20. A ray of light coming from the point $\left(2,2\sqrt{3}\right)$ is incident at an angle 30° on the line $x=1$ at the point A. The ray gets reflected on the line $x=1$ and meets x-axis at the point B. Then, the line AB passes through the point: **[JEE (Main) 2020]**

(1) $\left(3,-\frac{1}{\sqrt{3}}\right)$ (2) $\left(3,-\sqrt{3}\right)$ (3) $\left(4,-\frac{\sqrt{3}}{2}\right)$ (4) $\left(4,-\sqrt{3}\right)$

MSL059

21. Let the equation of the pair of lines, $y=px$ and $y=qx$, can be written as $(y-px)(y-qx)=0$. Then the equation of the pair of the angle bisectors of the lines $x^2-4xy-5y^2=0$ is : **[JEE (Main) 2021]**

(1) $x^2-3xy+y^2=0$ (2) $x^2+4xy-y^2=0$

(3) $x^2+3xy-y^2=0$ (4) $x^2-3xy-y^2=0$

MSL061

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22. If p and q are the lengths of the perpendiculars from the origin on the lines, $x\operatorname{cosec}\alpha - y\sec\alpha = k\cot 2\alpha$ and $x\sin\alpha + y\cos\alpha = k\sin 2\alpha$ respectively, then k^2 is equal to : **[JEE (Main) 2021]**

(1) $4p^2+q^2$ (2) $2p^2+q^2$ (3) p^2+2q^2 (4) p^2+4q^2

MSL062

23. The equations of the sides AB, BC and CA of a triangle ABC are $2x+y=0, x+py=15a$ and $x-y=3$ respectively. If its orthocentre is $(2,a)$, $-\frac{1}{2}<a<2$, then p is equal to

[JEE (Main) 2022]

MSL064

24. In an isosceles triangle ABC, the vertex A is (6, 1) and the equation of the base BC is $2x+y=4$. Let the point B lie on the line $x+3y=7$. If (α, β) is the centroid ΔABC, then $15(\alpha+\beta)$ is equal to: **[JEE (Main) 2022]**

(1) 39 (2) 41 (3) 51 (4) 63

MSL096

25. The equations of the sides AB, BC and CA of a triangle ABC are $2x+y=0$, $x+py=39$ and $x-y=3$ respectively and $P(2,3)$ is its circumcentre. Then which of the following is NOT true: **[JEE (Main) 2022]**

(1) $(AC)^2=9p$ (2) $(AC)^2+p^2=136$

(3) $32 <$ area $(\Delta ABC) < 36$ (4) $34 <$ area $(\Delta ABC) < 38$

MSL097

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26. Let the circumcentre of a triangle with vertices $A(a, 3)$, $B(b, 5)$ and $C(a, b)$, $ab > 0$ be $P(1, 1)$. If the line AP intersects the line BC at the point $Q(k_1, k_2)$, then $k_1 + k_2$ is equal to: **[JEE (Main) 2022]**

(1) 2 (2) $\frac{4}{7}$ (3) $\frac{2}{7}$ (4) 4

MSL098

27. Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2+11a+3\left(m_2^2+m_2^2\right)=220$. If one vertex of the square is $(10(\cos\alpha - \sin\alpha), 10(\sin\alpha + \cos\alpha))$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha - \sin\alpha)\, x + (\sin\alpha + \cos\alpha)\, y = 10$, then $72(\sin^4\alpha + \cos^4\alpha) + a^2 - 3a + 13$ is equal to: **[JEE (Main) 2022]**

(1) 119 (2) 128 (3) 145 (4) 155

MSL099

28. Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1: 3x - 4y + 12 = 0$, and $L_2: 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2, then $100(\alpha + \beta)$ is equal to

[JEE (Main) 2022]

(1) –14 (2) 42 (3) –22 (4) 14

MSL100

29. The distance between the two points A and A' which lie on $y = 2$ such that both the line segments AB and $A'B$ (where B is the point (2, 3)) subtend angle $\frac{\pi}{4}$ at the origin, is equal to :

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[JEE (Main) 2022]

(1) 10 (2) $\frac{48}{5}$ (3) $\frac{52}{5}$ (4) 3

MSL101

30. If the point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on the curve traced by the mid-points of the line segments of the lines $x\cos\theta + y\sin\theta = 7$, $\theta \in \left(0, \frac{\pi}{2}\right)$ between the co-ordinates axes, then α is equal to **[JEE (Main) 2023]**

(1) 7 (2) –7 (3) $-7\sqrt{3}$ (4) $7\sqrt{3}$

MSL103

31. A triangle is formed by X – axis, Y– axis and the line $3x + 4y = 60$. Then the number of points $P(a, b)$which lie strictly inside the triangle, where a is an integer and b is a multiple of a, is____.

[JEE (Main) 2023]

MSL104

32. A light ray emits from the origin making an angle 30° with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q, then the abscissa of Q is **[JEE (Main) 2023]**

(1) $\frac{2}{\left(\sqrt{3}-1\right)}$ (2) $\frac{2}{3+\sqrt{3}}$ (3) $\frac{2}{3-\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2\left(\sqrt{3}+1\right)}$

MSL105

For More Join: @IITJEE_Advanced

33. A straight line cuts off the intercepts $OA=a$ and $OB=b$ on the positive directions of x–axis and y–axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y–axis and the area of ΔOAB is $\frac{98}{3}\sqrt{3}$, then a^2-b^2 is equal to:

[JEE (Main) 2023]

(1) $\frac{392}{3}$ (2) 196 (3) $\frac{196}{3}$ (4) 98

MSL106

34. Let the equations of two adjacent sides of a parallelogram $ABCD$ be $2x-3y=-23$ and $5x+4y=23$. If the equation of its one diagonal AC is $3x+7y=23$ and the distance of A from the other diagonal is d, then $50\,d^2$ is equal to ____ . **[JEE (Main) 2023]**

MSL107

35. Let B and C be the two points on the line $y+x=0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y-2x=2$ such that ΔABC is an equilateral triangle. Then, the area of the ΔABC is **[JEE (Main) 2023]**

(1) $3\sqrt{3}$ (2) $2\sqrt{3}$ (3) $\frac{8}{\sqrt{3}}$ (4) $\frac{10}{\sqrt{3}}$

MSL108

36. Consider a triangle ABC having the vertices $A(1,2)$, $B(\alpha,\beta)$ and $C(\gamma,\delta)$ and angles $\angle ABC=\frac{\pi}{6}$ and $\angle BAC=\frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to .....

[JEE (Main) 2024]

MSL168

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37. The vertices of a triangle are $A(-1, 3)$, $B(-2, 2)$ and $C(3, -1)$. A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is : **[JEE (Main) 2024]**

(1) $x-y-\left(2+\sqrt{2}\right)=0$ (2) $-x+y-\left(2-\sqrt{2}\right)=0$

(3) $x+y-\left(2-\sqrt{2}\right)=0$ (4) $x+y+\left(2-\sqrt{2}\right)=0$

MSL169

38. Let $A(-1, 1)$ and $B(2, 3)$ be two points and P be a variable point above the line AB such that the area of ΔPAB is 10. If the locus of P is $ax + by = 15$, then $5a + 2b$ is: **[JEE (Main) 2024]**

(1) $-\frac{12}{5}$ (2) $-\frac{6}{5}$ (3) 4 (4) 6

MSL170

39. Let two straight lines drawn from the origin O intersect the line $3x + 4y = 12$ at the points P and Q such that ΔOPQ is an isosceles triangle and $\angle POQ = 90°$. If $l = OP^2 + PQ^2 + QO^2$, then the greatest integer less than or equal to l is : **[JEE (Main) 2024]**

(1) 44 (2) 48 (3) 46 (4) 42

MSL171

40. If the locus of the point, whose distances from the point (2, 1) and (1, 3) are in the ratio 5 : 4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to:

[JEE (Main) 2024]

(1) 5 (2) –27 (3) 37 (4) 437

MSL172

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EXERCISE - JEE (Advanced) PYQ

1. Consider the lines given by
$L_1 = x + 3y - 5 = 0$
$L_2 = 3x - ky - 1 = 0$
$L_3 = 5x + 2y - 12 = 0$ **[IIT JEE 2008]**

Match the statements/Expression in **Column-I** with the statements/Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in OMR.

Column-I	**Column-II**
(A) L_1, L_2, L_3 are concurrent, if	(P) $k = -9$
(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(Q) $k = -\frac{6}{5}$
(C) L_1, L_2, L_3 form a triangle, if	(R) $k = \frac{5}{6}$
(D) L_1, L_2, L_3 do not form a triangle, if	(S) $k = 5$

MSL083

2. A straight-line L through the point $(3, -2)$ is inclined at an angle $60°$ to the line $\sqrt{3}x + y = 1$. If L also intersect the x-axis, then the equation of L is **[IIT JEE 2011]**

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$ (B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$ (D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

MSL084

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3. For $a > b > c > 0$, the distance between $(1, 1)$ and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then **[JEE (Advanced) 2013]**

(A) $a + b - c > 0$ (B) $a - b + c < 0$ (C) $a - b + c > 0$ (D) $a + b - c < 0$

MSL085

4. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4$, is **[JEE (Advanced) 2014]**

MSL086

Question Stem for Question Nos. 5 and 6

Question Stem
Consider the lines L_1 and L_2 defined by
$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$
For a fixed constant λ, let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S, where the distance between R and S is $\sqrt{270}$.
Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'.

5. The value of λ^2 is _____. **[JEE (Advanced) 2021]**

MSL087

6. The value of D is _____. **[JEE (Advanced) 2021]**

MSL088

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	D	A	C	D	A	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	D	D	B	B	B	C	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	A	A	B	B	B	D	A	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C	B,D	B,C	C,D	B,C	B,C,D	A,B,C	A,D	A,B,C,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	B,D	A,C,D	A,D	B	D	A	C	A,B,C	C	
Que.	20									
Ans.	A→R; B→P; C→Q; D→T									

EXERCISE (S)

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1. 5 **2.** 3 **3.** 91 **4.** 47 **5.** 6
6. 8 **7.** 6 **8.** 24 **9.** 89 **10.** 2

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	2	2	4	3	1	3	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1 or 2	3	5	3	2	4	4	30	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	3	3	4	2	2	4	3	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	31	2	1	529	3	14	3	1	3	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6
Ans.	A→S; B→P,Q; C→R; D→ P,Q,S	B	A,C	6	9.00	77.17

For More Join: @IITJEE_Advanced